

PRATHAMESH PRADIP NIBANDHE

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Summary

MCA student with an engineering background and hands-on experience applying Python and large language models (Gemini, Groq) to build AI-driven automation systems. Experience translating natural-language prompts into functional code/firmware, integrating AI APIs into production pipelines, and collaborating on live R&D projects spanning software and embedded systems. National hackathon winner in an Agentic AI competition, with a strong foundation in coding, data-driven problem solving, and rapid prototyping.

Work Experience

Developer Intern — InFranex

January 2026 – Present

- Building integrated web, IoT, and software systems that process real-time data from 10+ embedded devices, working alongside engineers on live production pipelines.
- Automating deployment workflows using Python-based tooling, reducing manual effort by ~40%.
- Contributing to R&D in robotics and system architecture, including ROS 2-based automation pipelines.

Hardware & Network Trainee — SS Computers

March 2024 – October 2024

- Monitored client servers and applications and conducted security audits, reducing downtime by 15% while ensuring compliance with IT standards.

Projects

Terrain Segmentation for Autonomous Vehicles — Duality AI Hackathon

2026

- Built a custom lightweight encoder-decoder (mini U-Net) segmentation model from scratch no pretrained weights to help an unmanned autonomous vehicle classify terrain (land, sky, river, rocky plains) from a labeled desert-image dataset; team of 3.
- Owned model training: implemented the training loop in PyTorch (CrossEntropyLoss, Adam) with torchvision/PIL/matplotlib, iterating on architecture and hyperparameters to optimize segmentation quality against the team's IoU benchmark.

GitHub: github.com/prathamesh-coding/yolo-Pune-Edition-DualityAi

OMNI-X — AI for Hardware

March 2026 – Present

- Built a hybrid Web IDE and CLI that converts natural-language prompts into fully functional ESP32 firmware using Python (Flask), PlatformIO, and LLMs (Gemini / Groq).
- Automated the full IoT lifecycle AI code generation, compilation, and wireless OTA updates eliminating manual firmware flashing.
- Currently powers firmware deployment for a modular farming robot with interchangeable soil-sensing, planting, and movement modules.

PRITH-V — Modular Farming Robot

October 2025 – Present

- Developing a modular robotics system with custom chassis, motor control, and 3D-prototyped interchangeable modules (ROS 2, C++) to adapt tools for 5+ distinct farming tasks, integrated with the OMNI-X AI firmware pipeline.

Achievements & Competitions

- 1st Rank — Agentic AI Hackathon, PES Modern College of Engineering
- 2nd Rank — Odoo x VIT Hackathon (May 2026)
- 4th Rank / Top 10 Finalist — HACKOVERFLOW 4.0 2026 (National Hackathon)

Technical Skills

Languages: Python, C++, C, JavaScript, Java, SQL

AI / ML: Machine Learning, Deep Learning, PyTorch, torchvision, CNN, U-Net (image segmentation), Gemini API, Groq API, prompt-to-code generation

Tools & Frameworks: Flask, ROS 2, PlatformIO, React.js, Node.js, Git/GitHub

Databases: MongoDB, MySQL, Firebase, Supabase

Education

MCA (Master of Computer Applications)

2025 – 2027

PES Modern College of Engineering, Pune — 6.55 CGPA

B.Sc. Computer Science

2022 – 2025

PVG College of Science and Commerce — 8.01 CGPA